
Windows PowerShell Basics Lab

Course: Operating Systems and You: Becoming a Power User - Module 1 – Windows CLI Fundamentals

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1. Introduction

This lab aims to introduce the use of the Windows Command Line Interface (CLI) through PowerShell.

During the lab, basic file and directory management tasks were performed using fundamental PowerShell commands, allowing the understanding of absolute and relative paths, directory navigation, file handling, and basic automation from the terminal.

The use of wildcards, command history, and advanced parameters such as -Recurse, -Verbose, and -Force was also practiced.

2. Environment Used

The lab was performed using the following environment:

- Virtualized Windows 11
- PowerShell
- macOS Host
- Windows Virtual Machine

3. Directory Navigation

Directory navigation was practiced using relative and absolute paths.

Commands Used

```
pwd
ls
cd
cd ..
cd ~
cd ~\Desktop
```

Concepts Applied

- Directory navigation
- Relative and absolute paths
- Use of the user home directory (~)
- Use of the parent directory (..)

4. Creating Directories and Files

Folders and files were created from PowerShell to practice basic file system administration.

Commands Used

```
mkdir test_folder  
mkdir backup  
mkdir Pics
```

```
New-Item file1.txt  
New-Item file2.txt  
New-Item image1.jpg
```

Concepts Applied

- Creating directories with mkdir
- Creating files with New-Item
- Basic file organization

Figure 1 – Created Directory and File Structure

```
PS C:\Users\Flor\Desktop> ls  
  
Directorio: C:\Users\Flor\Desktop  
  
Mode                LastWriteTime         Length Name  
----                -  
d-----          26/05/2026    0:44          backup  
d-----          26/05/2026   22:03           Pics  
d-----          26/05/2026   22:05       test_folder  
  
PS C:\Users\Flor\Desktop> cd test_folder  
PS C:\Users\Flor\Desktop\test_folder> ls  
  
Directorio: C:\Users\Flor\Desktop\test_folder  
  
Mode                LastWriteTime         Length Name  
----                -  
-a-----          26/05/2026    22:04          0 file1.txt  
-a-----          26/05/2026    22:04          0 file2.txt  
-a-----          26/05/2026    22:05          0 image1.jpg  
  
PS C:\Users\Flor\Desktop\test_folder> |
```

5. Copying Files and Directories

Practices were performed involving copying individual files, multiple files using wildcards, and complete folders.

Commands Used

```
cp *.txt ../backup
cp *.jpg ../Pics
cp test_folder backup -Recurse
```

Concepts Applied

- Use of wildcards (*)
- Copying multiple files
- Copying complete directories using -Recurse

Verification of Copied Files

```
ls ../backup
ls ../Pics
```

Figure 2 – Copying Files Using Wildcard

```
PS C:\Users\Flor\desktop> cd test_folder
PS C:\Users\Flor\desktop\test_folder> ls

Directorio: C:\Users\Flor\desktop\test_folder

Mode                LastWriteTime         Length Name
----                -
-a-----         26/05/2026      22:04             0 file1.txt
-a-----         26/05/2026      22:04             0 file2.txt
-a-----         26/05/2026      22:05             0 image1.jpg

PS C:\Users\Flor\desktop\test_folder> cp *.txt ../backup
PS C:\Users\Flor\desktop\test_folder> ls ../backup

Directorio: C:\Users\Flor\desktop\backup

Mode                LastWriteTime         Length Name
----                -
d-----         25/05/2026      23:56          test_folder
-a-----         26/05/2026      22:04             0 file1.txt
-a-----         26/05/2026      22:04             0 file2.txt
```

6. Moving and Renaming

Practices involving moving and renaming files and directories were performed.

Commands Used

```
mv file1.txt documento.txt
mv documento.txt ../backup
mv test_folder lab_folder
```

Concepts Applied

- Renaming files
- Moving files between folders
- Difference between moving and renaming

Figure 3 – Moving and Renaming Files

```
Directorio: C:\Users\Flor\desktop\test_folder

Mode                LastWriteTime         Length Name
----                -
-a-----         26/05/2026    22:04             0 file1.txt
-a-----         26/05/2026    22:04             0 file2.txt
-a-----         26/05/2026    22:05             0 image1.jpg

PS C:\Users\Flor\desktop\test_folder> mv file1.txt documento.txt
PS C:\Users\Flor\desktop\test_folder> ls

Directorio: C:\Users\Flor\desktop\test_folder

Mode                LastWriteTime         Length Name
----                -
-a-----         26/05/2026    22:04             0 documento.txt
-a-----         26/05/2026    22:04             0 file2.txt
-a-----         26/05/2026    22:05             0 image1.jpg
```

7. Removing Files and Directories

File and directory deletion was practiced using PowerShell.

Commands Used

```
rm file2.txt
rm test_folder -Recurse
rm image1.jpg -Force
```

Concepts Applied

- Removing files using rm
- Recursive deletion with -Recurse
- Use of -Force
- Difference between CLI deletion and the recycle bin

Figure 4 – Removing Directories Using rm -Recurse

```
PS C:\Users\Flor\desktop> ls

Directorio: C:\Users\Flor\desktop

Mode                LastWriteTime         Length Name
----                -
d-----          26/05/2026   0:44             backup
d-----          26/05/2026  22:03             Pics
d-----          26/05/2026  22:51          test_folder

PS C:\Users\Flor\desktop> rm test_folder -Recurse
PS C:\Users\Flor\desktop> ls

Directorio: C:\Users\Flor\desktop

Mode                LastWriteTime         Length Name
----                -
d-----          26/05/2026   0:44             backup
d-----          26/05/2026  22:03             Pics
```

8. History and Productivity

Built-in PowerShell tools were used to improve productivity and workflow speed.

Commands Used

history
clear

Functions Used

- Command history
- Navigation using arrow keys ↑ ↓
- TAB autocomplete
- Screen clearing using clear

9. Tests Performed

During the lab, the following tasks were successfully validated:

- Directory navigation
- Use of relative and absolute paths
- Creation of files and folders
- Copying files using wildcards
- Copying complete folders
- Moving and renaming files
- Removing files and directories
- Content verification using ls

10. Analysis

The lab made it possible to understand how PowerShell manages files and directories through CLI commands.

One of the most important concepts was the difference between absolute and relative paths, especially when working with .., ~, and folders located at different levels of the file system.

It was also observed that PowerShell, in many cases, does not display messages when a command executes successfully, making it important to verify results using commands such as ls.

The use of wildcards allowed repetitive tasks to be automated, while parameters such as -Recurse and -Force demonstrated how PowerShell can perform more advanced administrative operations.

11. Conclusion

This lab helped develop a solid foundation in the use of PowerShell for basic file and directory management in Windows.

Fundamental CLI concepts were incorporated that will later be used in technical support, systems administration, and automation tasks.

Using terminal commands allows tasks to be completed faster, more precisely, and more efficiently compared to traditional graphical interfaces.

In addition, this lab served as a practical introduction to the daily use of administration tools commonly found in real IT environments.